

Truncated Noncharacteristic Deformation Lemma

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1 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(2) \quad E_\infty^\bullet: \cdots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \cdots.$$

Denote by

$$(3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

Proposition 1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^\bullet))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

ML procedure for sheaves

Proposition 3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathbf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

2 Noncharacteristic Deformation Lemma

Proposition 4. Let X be a Hausdorff space. Let $F \in \mathbf{D}^+(\mathbf{k}_X)$, and $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . Let $k \in \mathbf{Z}$. If s is a real number, set $Z_s := \bigcap_{t > s} \overline{U_t} \setminus U_s$. We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t} \setminus U_s \cap \text{supp } F$ is compact.
- (iii) For all $s \in \mathbf{R}$ and all $x \in Z_s \setminus U_s$, we have

$$(H_{X \setminus U_s}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For all pairs (s, t) with $s < t$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Example 5. Let us give examples of $(U_t)_t$ satisfying the condition iv, which is not contained in the original version of the proposition ([KS90, Proposition 2.7.2]). In all the examples below, X is $\mathbf{R}^2$.

- (i) For a real number t , define the open set U_t by

$$U_t := \begin{cases} \{(x, y) \in X; \|(x, y)\| < t\} & (t > 0), \\ \emptyset & (t \leq 0). \end{cases}$$

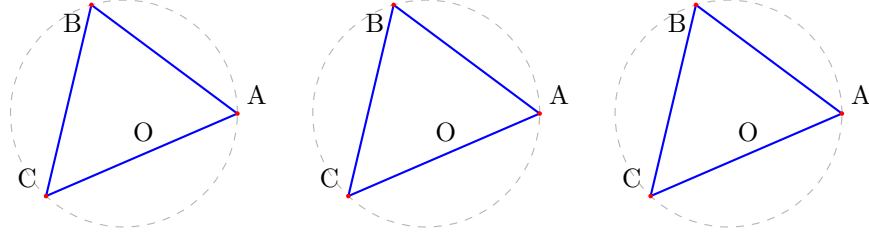


Figure 1: Examples of deformation families.

- (ii) For a real number t , define the function $\varphi_t: \mathbf{R} \rightarrow \mathbf{R}$ as follows. If $t \leq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) := \begin{cases} \sqrt{t^2 - x^2} & (|x| < t), \\ 0 & (|x| \geq t). \end{cases}$$

Then define the open set U_t by

$$U_t := \{(x, y) \in X; y < \varphi_t(x)\}.$$

- (iii) For a real number t , define the open set U_t as follows. If $t \leq 0$, set

$$U_t := \{(x, y) \in X; y < 0\}.$$

If $t > 0$, set

$$U_t := \left\{ (x, y) \in X; \begin{array}{ll} \text{If } |x| < \frac{1}{t}, & y < t. \\ \text{Otherwise,} & y < \frac{1}{|x|}. \end{array} \right\}.$$

These families are visualized as Figure 1.

Proof. Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_s; F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r; F). \end{aligned}$$

Let us prove, for all $s \in \mathbf{R}$, $(a)_s^j$ for $j < k$ and $(a')_s^k$. We may assume that $\overline{U_t} \setminus \overline{U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X \setminus U_s}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_s}(F) \xrightarrow{+1}.$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i_s: Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\cdot)|_{Z_s \cup U_s} = i_s^{-1}$$

to the d.t. above, we have

$$\mathrm{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s} \rightarrow F|_{Z_s \cup U_s} \rightarrow \mathrm{R}\Gamma_{U_s}(F)|_{Z_s \cup U_s} \xrightarrow{+1}.$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s \cup U_s; \cdot)$, we have

$$(4) \quad \begin{array}{ccc} & \mathrm{R}\Gamma(Z_s \cup U_s; \mathrm{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) & \\ \nearrow^{+1} & & \searrow \\ \mathrm{R}\Gamma(U_s; F) & \longleftarrow & \mathrm{R}\Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}). \end{array}$$

By hypotheses (iii) and the definition of local cohomology, it follows that

$$H_{X \setminus U_s}^j(F)|_{Z_s \cup U_s} = 0 \quad \text{for any } j \leq k,$$

and then

$$H^j(Z_s \cup U_s; \mathrm{R}\Gamma_{X \setminus U_s}(F)|_{Z_s \cup U_s}) = 0 \quad \text{for any } j \leq k.$$

Thus we have

$$(5) \quad H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for any } j < k,$$

$$(6) \quad H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \hookrightarrow H^k(U_s; F).$$

Therefore, if we show Lemma 6 below, we have

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows.

Lemma 6. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}).$$

Proof of Lemma 6 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$ there exists an isomorphism

$$(7) \quad \varinjlim_{t>s} \Gamma(U_t; F) \cong \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}).$$

Consider the following diagram.

$$\begin{array}{ccc}
\varinjlim_{t>s} \Gamma(U_t; F) & & \\
\downarrow & \searrow & \\
\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) & \longrightarrow & \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \\
\downarrow & & \downarrow \\
\varinjlim_{U \supset Z_s} \Gamma(U; F) & \longrightarrow & \Gamma(Z_s; F|_{Z_s})
\end{array}$$

Since $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U \supset Z_s}$, we have

$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F),$$

so it is enough to show that

$$\phi: \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \rightarrow \Gamma(Z_s \cup U_s; F)$$

is isomorphic. Since X is Hausdorff and Z_s is compact, the natural morphism

$$\psi: \varinjlim_{U \supset Z_s} \Gamma(U; F) \rightarrow \Gamma(Z_s; F)$$

is isomorphic (see [KS90, Proposition 2.5.1]). Since U_s is open, we have another isomorphism

$$\psi': \varinjlim_{V \supset U_s} \Gamma(V; F) \rightarrow \Gamma(U_s; F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ' .

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s . More precisely, there exists a family $(\alpha^U)_U \in \varinjlim_{U \supset Z_s} \Gamma(U; F)$ such that

$$\alpha^U|_{Z_s} = \sigma|_{Z_s}$$

for all U . Similarly, there exists a family $(\beta^V)_V \in \varinjlim_{V \supset U_s} \Gamma(V; F)$ such that

$$\beta^V|_{U_s} = \sigma|_{U_s}$$

for all V . These $(\alpha^U)_U$ and $(\beta^V)_V$ satisfy

$$\alpha^U|_{U \cap U_s} = \beta^V|_{U \cap U_s}.$$

Therefore, by the definition of a sheaf, there exists a unique section $\gamma \in \Gamma(U \cup U_s; F)$ such that

$$\gamma|_U = \alpha^U, \quad \gamma|_{U_s} = \beta^{U_s}.$$

For this γ , we have

$$\gamma|_{Z_s \cup U_s} = \sigma$$

and it follows that

$$\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \rightarrow \Gamma(Z_s \cup U_s; F)$$

is an isomorphism.

Next Let $F \in D^+(\mathbf{k}_X)$. Let $I^\bullet$ be a complex of flabby sheaves quasi-isomorphic to F . Then it follows from (7) that

$$\begin{aligned} \varinjlim_{t > s} H^j(U_t; F) &\cong \varinjlim_{t > s} H^j \Gamma(U_t; I^\bullet) \\ &\cong H^j \left(\varinjlim_{t > s} \Gamma(U_t; I^\bullet) \right) \\ &\cong H^j \Gamma(Z_s \cup U_s; I^\bullet|_{Z_s \cup U_s}) \\ &\cong H^j(Z_s \cup U_s; F|_{Z_s \cup U_s}). \end{aligned}$$

This completes the proof. $\square$

Let us come back to the proof of Proposition 4. We prove $(b)_s^j$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j . Since $F \in D^+(\mathbf{Z}_X)$, we have $(b)_s^j$ for $j \ll 0$ and $s \in \mathbf{R}$. Assume $(b)_s^j$ is proved for all $j < j_0 (< k)$ and $s \in \mathbf{R}$. Since we have $(a)_s^j$ and $(b)_s^j$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $(H^{j_0-1}(U_{s-1/n}))_{n \in \mathbf{N}}$. This system satisfies ML condition, so, by applying Proposition 3 (iii), we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j(U_{s-1/n}; F).$$

Since $(s-1/n)_n$ is a cofinal family of $(t)_{s > t}$, we have $(b)_s^{j_0}$. Then by induction on j , we have $(b)_s^j$ for $j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 3 to the projective system $(H^j(U_n; F))_n$, we obtain

$$\begin{aligned} H^j \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) &\cong H^j \left(\bigcup_{n \in \mathbf{N}} U_n; F \right) \\ &\xrightarrow{\sim} \varprojlim_n H^j(U_n; F) \end{aligned}$$

and for each n , we have by Kashiwara's lemma that

$$H^j(U_n; F) \cong H^j(U_s; F)$$

for $n \geq s$.

Finally we replace $(a)_s^j$ by $(a')_s^k$ for $j = k$, and obtain

$$H^k \left(\bigcup_{s \in \mathbf{R}} U_s; F \right) \hookrightarrow H^k(U_s; F).$$

This completes the proof. $\square$

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References

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